DRUGS · HEAD AND NECK ONCOLOGY

Bevacizumab

Bevacizumab is a monoclonal antibody that binds vascular endothelial growth factor (VEGF) and suppresses new vessel formation. Given intravenously, it has transformed the management of severe recurrent respiratory papillomatosis, reducing the frequency of surgery, and it reduces bleeding and transfusion needs in hereditary haemorrhagic telangiectasia with severe epistaxis or gastrointestinal bleeding. It carries the class risks of anti-VEGF drugs: hypertension, proteinuria, bleeding, thrombosis and poor wound healing.

1 CHAPTER ONE At a glance

The drug on one page.

1.1 Summary

BEVACIZUMAB

	SUMMARY
Class	Anti-VEGF monoclonal antibody
How it works	Binds VEGF-A, preventing receptor activation; reduces angiogenesis and vascular permeability
ENT uses	Severe recurrent respiratory papillomatosis (systemic, as adjuvant to surgery); hereditary haemorrhagic telangiectasia with severe or transfusion-dependent bleeding; intralesional use in papillomatosis has been reported
Dose	Regimens vary: commonly 5 to 10 mg/kg intravenously every 2 to 3 weeks for an induction phase, then extended intervals for maintenance (specialist protocols)
Avoid	Recent major surgery (wait at least 4 weeks; stop before elective surgery), uncontrolled hypertension, active bleeding, pregnancy

2 CHAPTER TWO Using it in ENT

Practice.

2.1 Recurrent respiratory papillomatosis

For children and adults with frequent surgery or pulmonary spread, systemic bevacizumab with debulking surgery lengthens intervals between procedures and can induce remission in many case series; it is used under a multidisciplinary team.

2.2 Hereditary haemorrhagic telangiectasia

For severe epistaxis despite local treatments (laser, coblation, septodermoplasty, Young's procedure), systemic bevacizumab reduces bleeding and transfusion needs. Topical intranasal bevacizumab did not help in trials. Tranexamic acid and iron replacement continue.

3 CHAPTER THREE Safety

What goes wrong.

3.1 Adverse effects

Hypertension, **proteinuria**, epistaxis or bleeding, arterial and venous thrombosis, delayed wound healing and wound dehiscence, gastrointestinal perforation, osteonecrosis of the jaw, posterior reversible encephalopathy syndrome.

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- Its role in **papillomatosis** and **HHT**.
- Monitoring (blood pressure, urine protein) and the surgery interval.

4.2 Common pitfalls

COMMON PITFALLS

- Operating on a patient who had bevacizumab within the last few weeks.

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